The Science Behind Running

A Data Science Perspective

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What Limits Human Endurance?

- “Every morning in Africa, a gazelle wakes up; it knows it must outrun the fastest lion or it will be killed. Every morning in Africa, a lion wakes up; it knows it must run faster than the slowest gazelle, or it will starve. It doesn’t matter whether you’re the lion or a gazelle—when the sun comes up, you’d better be running (African Proverb).”
- World records continue improving by fractions of a second
- Raises the question: **What defines the limits of human endurance?**
- Key factors: VO2max, lactate threshold, running economy, metabolism, neurological factors, genetics, mathematical models

Early Origins of Running

- 300,000 years ago: Homo sapiens evolve endurance running abilities
- Key features: sweat glands, Achilles tendon, compound knee, gluteus maximus
- Running tied to hunting and scavenging
- Earliest competitive races: 1829 BCE
- Running in ancient cultures: Greece, Egypt, Asia, East African Rift

Origins of the Olympics

- Olympic Games began in 776 BCE
- Ran continuously until 393 AD
- Running events central to Greek culture

The First Marathon

- “Only the disciplined ones in life are free (Eliud Kipchoge).”
- 1896 Olympics: First organized marathon
- Inspired by legend of Pheidippides (490 BCE)
- An ancient Greek messenger ran from Marathon to Athens, collapsed, died
- Distance solidified later at 26.2 miles

Kipchoge's Historic Run

- October 12, 2019, Vienna
- Eliud Kipchoge runs **1:59:40.2**
- First human under 2 hours in the marathon
- The greatest marathoner ever (the Goat)

Roger Bannister's Mile Barrier

- "The man who can drive himself further once the effort gets painful is the man who will win (Sir Roger Banister)."
- May 6, 1954: Bannister runs **3:59.4**
- First person to break the four-minute mile
- Today: Over 1,600 athletes have broken it
- Current world record: Hicham El Guerrouj, **3:43.13** (1999)

How the Body Produces Energy

Aerobic:

- Uses oxygen
- Breaks down carbs, fats, amino acids
- Produces ATP efficiently

Anaerobic:

- High intensity
- No oxygen
- Produces lactic acid

Endurance running relies primarily on **aerobic capacity**

What Makes a Great Endurance Runner?

- **VO2max:** upper limit of oxygen utilization
- **Lactate threshold:** determines sustainable intensity
- **Running economy:** efficiency of energy use
- **Neurological factors:** pain tolerance, muscle recruitment
- **Genetics:** influences efficiency and fatigue resistance
- **Training Quality and Volume:** Mileage per week, Quality of workouts, Long-term progression, Proper recovery
- **Strength & Biomechanics:** Core strength and stability, Hip strength, Stride mechanics, Ground contact time, Cadence
- **Nutrition & Fueling:** Carbohydrates, Amino Acids, Fatty Acids, Nucleic Acids, Glycogen storage, Hydration, Electrolytes, Iron levels
- **Body Composition:** Low body-fat percentage, High muscle efficiency, Distance running favors lean, efficient bodies.
- **Psychological Factors:** Motivation, Mental toughness, Racing strategy, Stress management, The “central governor” theory says the brain limits performance before the body does.
- **Technology & Equipment:** Super shoes (carbon plates, foam technology), Lightweight clothing, GPS pacing
- **Environmental Factors:** Altitude, Heat, Humidity, Tailwind/headwind

Human Limits

- Many variables interact to determine the true limits of running performance
- Men's mile world record has improved **30 seconds** over the past 100 years
- Theoretical physics limit: **~5.3×10^{-6} s**
- If Usain Bolt held his 100m pace for a mile: **2:34.18**
- Rudisha's 800m pace suggests a lower limit near **3:21.82**
- Current world record: **3:43.13**
- Only about **1,600 runners** have ever broken the **4:00 mile**

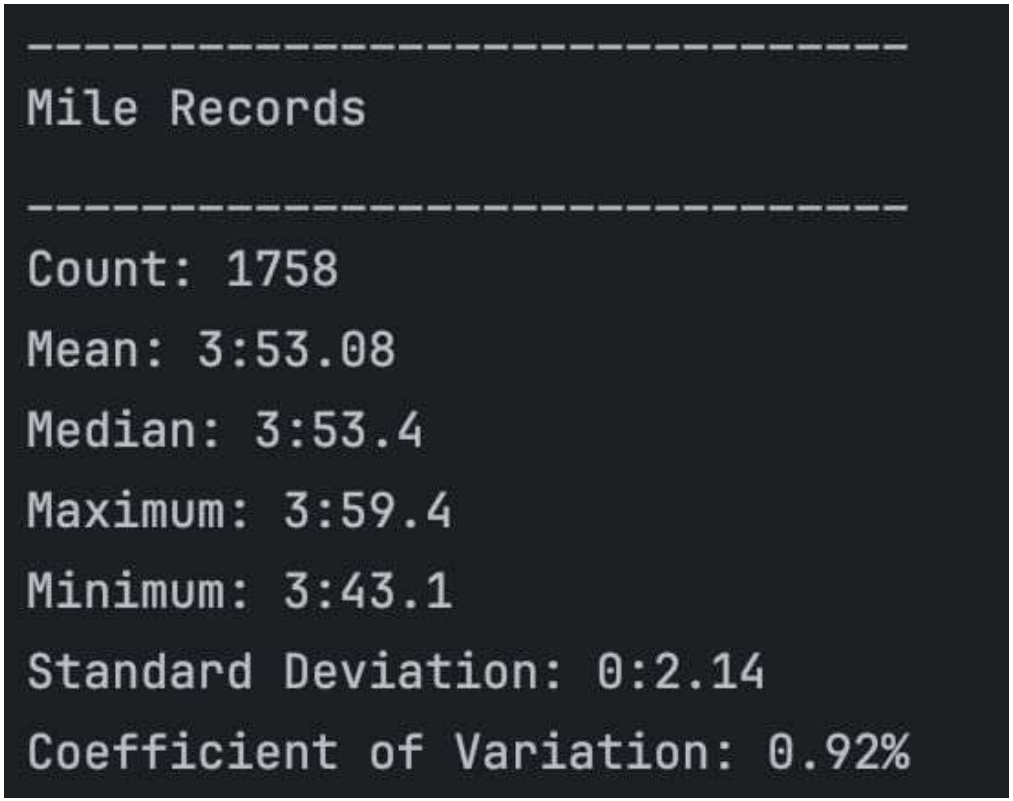
World Records

Rank	Mark	Competitor	DOB	Nat	Date	Event	Wind	Sex
1	9.58	Usain BOLT	8/21/86	JAM	8/16/09	100 Metres	0.9	male
1	12.8	Aries MERRITT	7/24/85	USA	9/7/12	110 Metres Hurdles	0.3	male
1	19.19	Usain BOLT	8/21/86	JAM	8/20/09	200 Metres	-0.3	male
1	30.69	Letsile Tebogo	6/7/03	BOT	2/17/24	300 Metres	nan	male
1	43.03	Wayde VAN NIEKERK	7/15/92	RSA	8/14/16	400 Metres	nan	male
1	45.94	Karsten WARHOLM	2/28/96	NOR	8/3/21	400 Metres Hurdles	nan	male
1	01:41.0	David RUDISHA	12/17/88	KEN	8/9/12	800 Metres	nan	male
1	02:11.9	Noah NGENY	11/2/78	KEN	9/5/99	1000 Metres	nan	male
1	03:26.0	Hicham EL GUERROUJ	9/14/74	MAR	7/14/98	1500 Metres	nan	male
1	03:43.1	Hicham EL GUERROUJ	9/14/74	MAR	7/7/99	One Mile	nan	male

All-Time Mile Times Ran

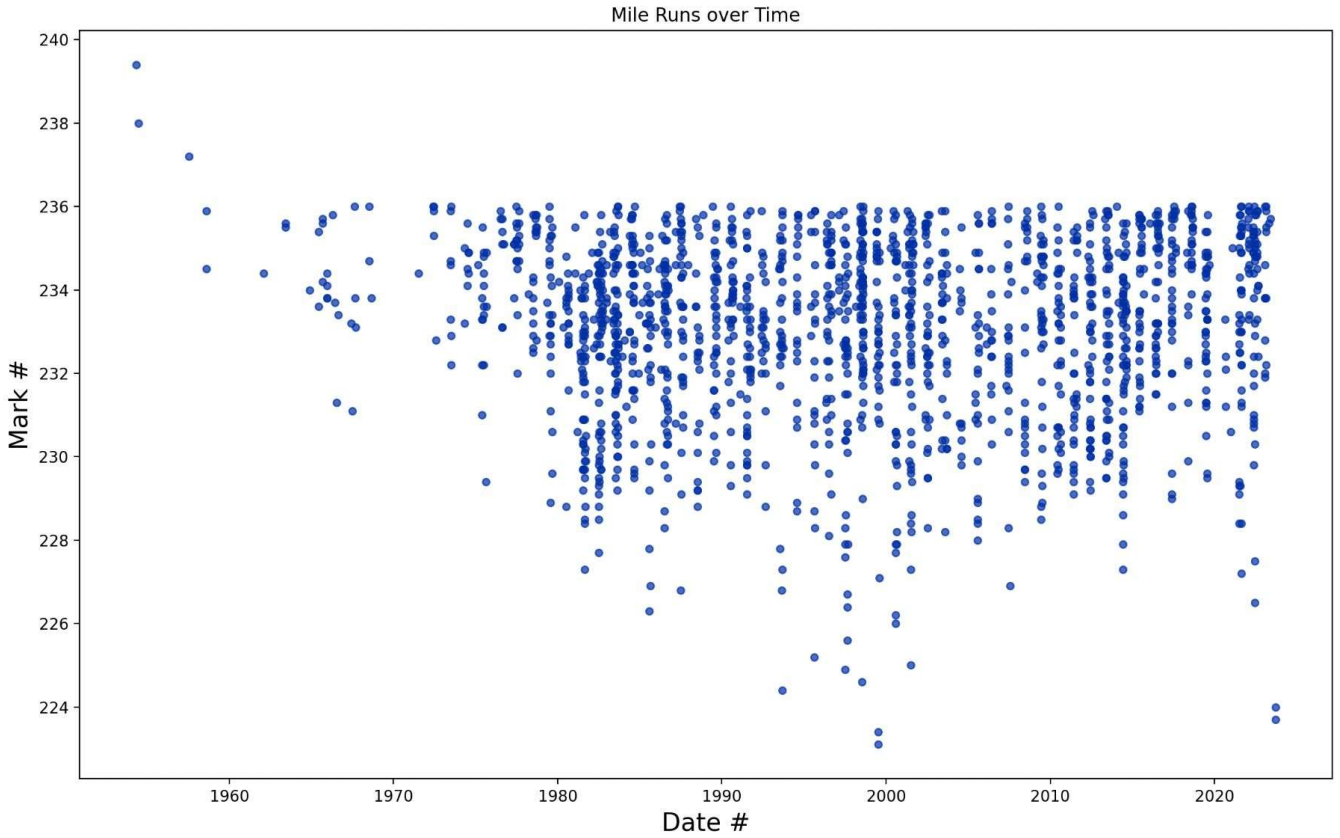
Competitor	Rank	Mark	Mark #	DOB	Nat	Venue	Date	Results Score
Hicham EL GUERROUJ	1	03:43.1	223.1	14-Sep-74	MAR	Bislett Stadion, Oslo (NOR)	12-Jul-02	1153
Noah NGENY	2	03:43.4	223.4	2-Nov-78	KEN	Crystal Palace, London (GBR)	13-Jul-01	1141
Jakob INGEBRIGTSEN	3	03:43.7	223.7	19-Sep-00	NOR	Bislett Stadion, Oslo (NOR)	13-Jun-19	1162
Yared NUGUSE	4	03:44.0	224	1-Jun-99	USA	Cardinal Gibbons High School, Raleigh, NC (USA)	16-Sep-23	1144
Noureddine MORCELI	5	03:44.4	224.4	28-Feb-70	ALG	Athina (GRE)	1-Sep-95	1134
Steve CRAM	13	03:46.3	226.3	14-Oct-60	GBR	Athina (GRE)	1-Jul-80	1149
Daniel KOMEN	14	03:46.4	226.4	17-May-76	KEN	Eugene, OR (USA)	21-Jul-98	1140
Vénuste NIYONGABO	16	03:46.7	226.7	9-Dec-73	BDI	Berlin (GER)	22-Jul-94	1126
Saïd AOUITA	17	03:46.8	226.8	2-Nov-59	MAR	Berlin (GER)	1-Sep-83	1144
Alan WEBB	19	03:46.9	226.9	13-Jan-83	USA	Brasschaat (BEL)	19-Jun-04	1124

Mile Records Data Analysis



The dataset contains all times of 3:56 or faster, except for Roger Bannister and the two previous world record holders.

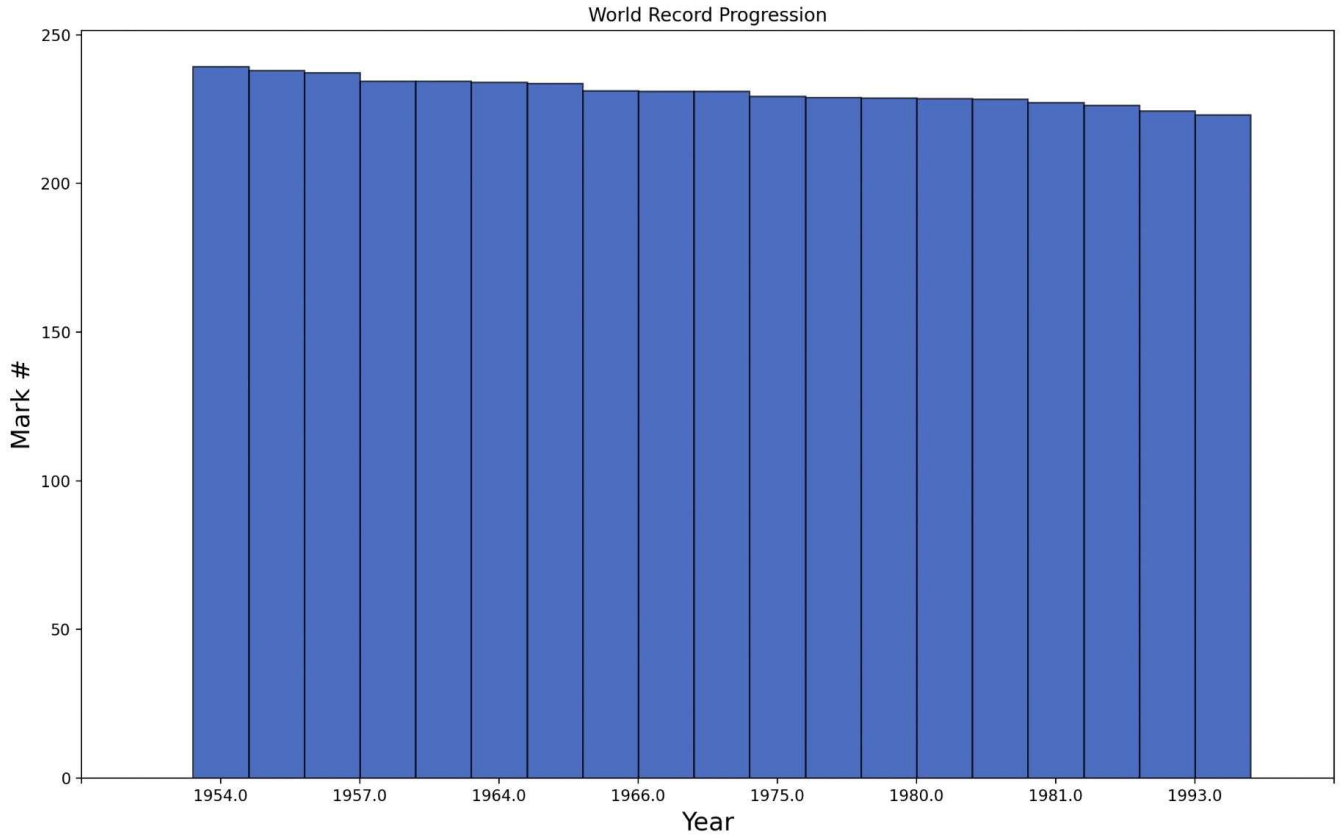
Mile Runs over Time



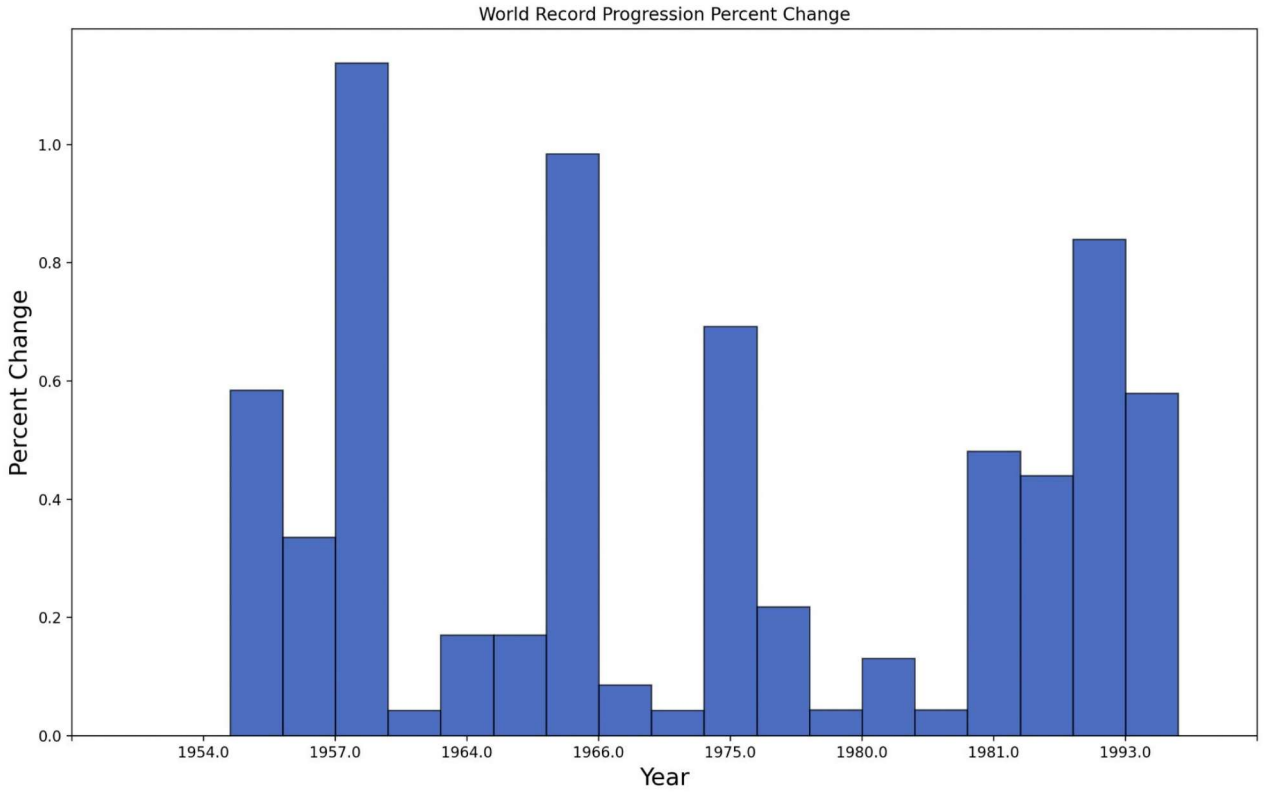
World Record Progression

Competitor	Year	Mark #
Roger BANNISTER	1954	239.4
John LANDY	1954	238
Derek IBBOTSON	1957	237.2
Herb ELLIOTT	1958	234.5
Peter SNELL	1962	234.4
Peter SNELL	1964	234
Michel JAZY	1965	233.6
Jim RYUN	1966	231.3
Jim RYUN	1967	231.1
Filbert BAYI	1975	231
John WALKER	1975	229.4
Sebastian COE	1979	228.9
Steve OVETT	1980	228.8
Sebastian COE	1981	228.5
Steve OVETT	1981	228.4
Sebastian COE	1981	227.3
Steve CRAM	1985	226.3
Noureddine MORCELI	1993	224.4
Hicham EL GUERROUJ	1999	223.1

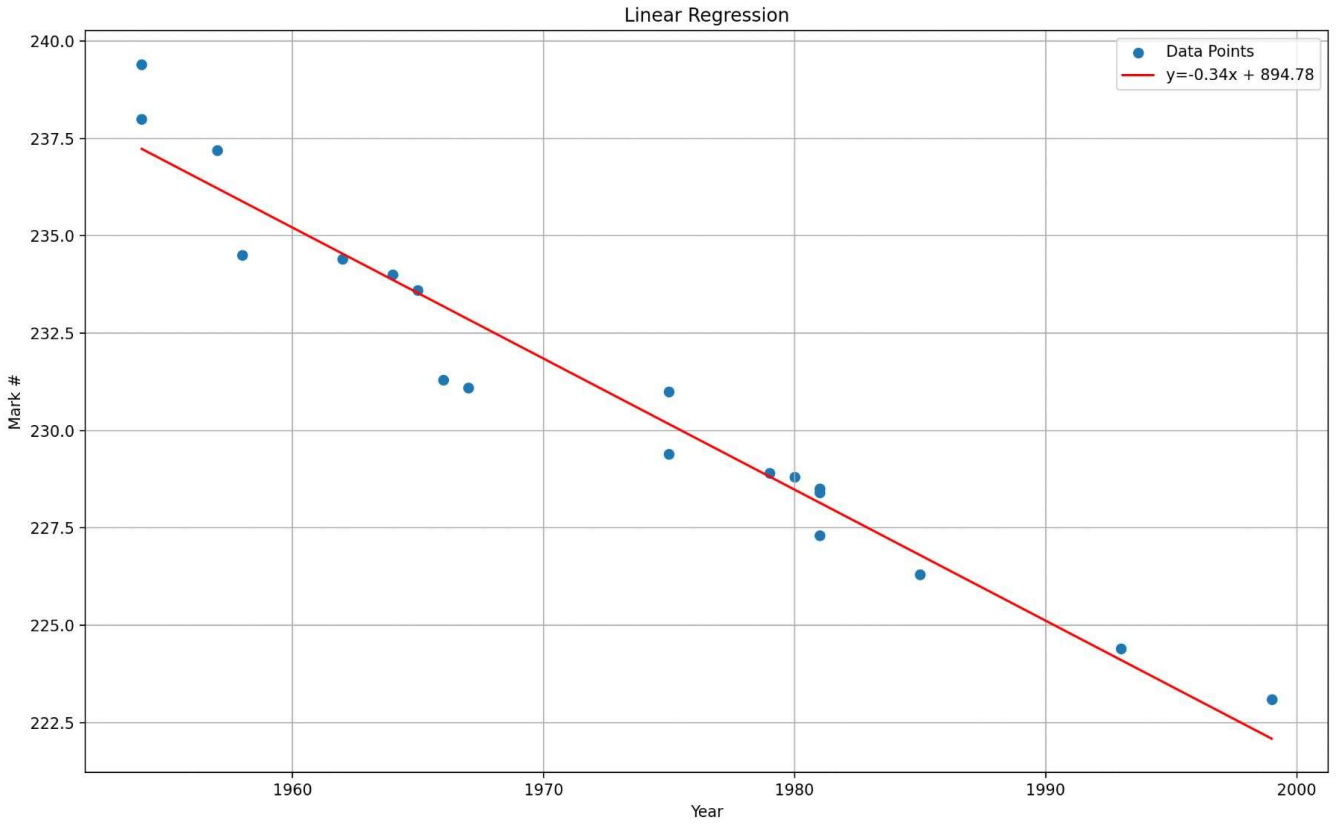
World Record Progression



World Record Progression Percent Change

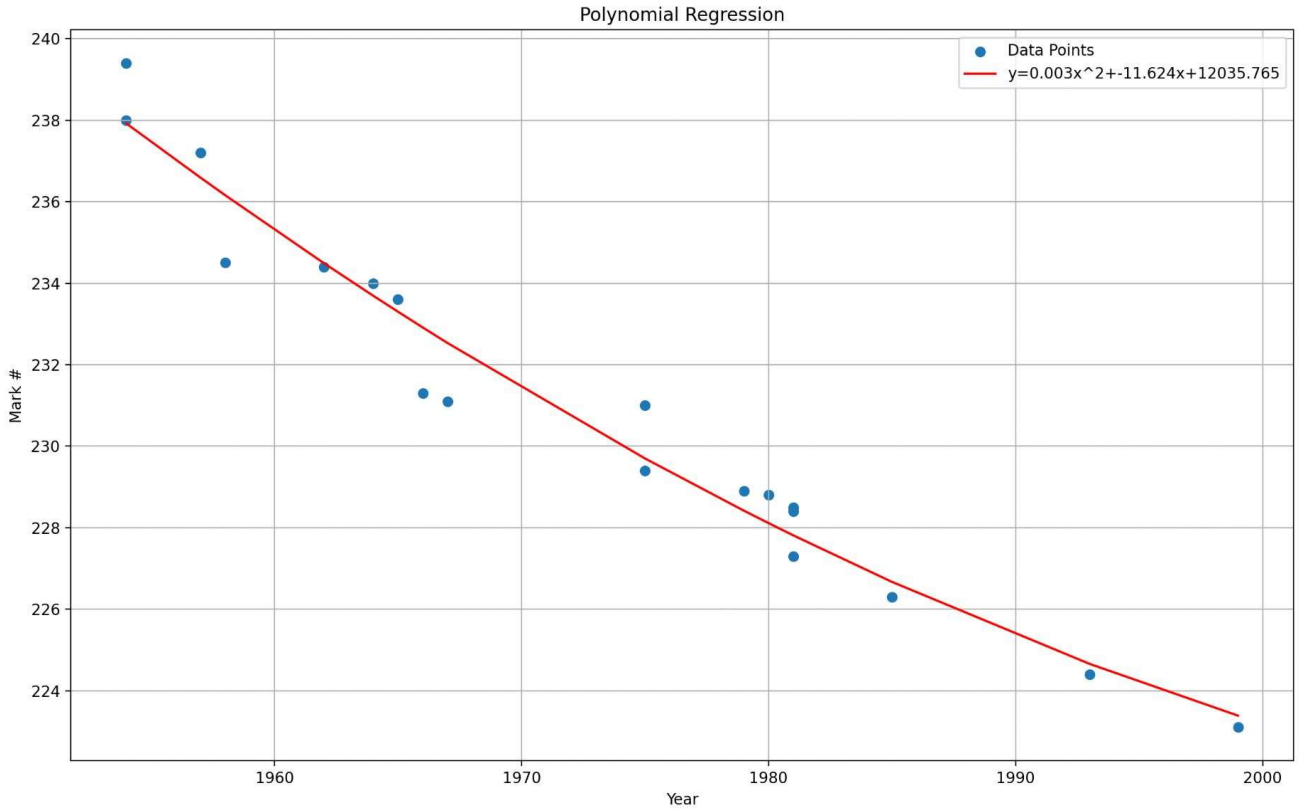


Linear Regression World Records



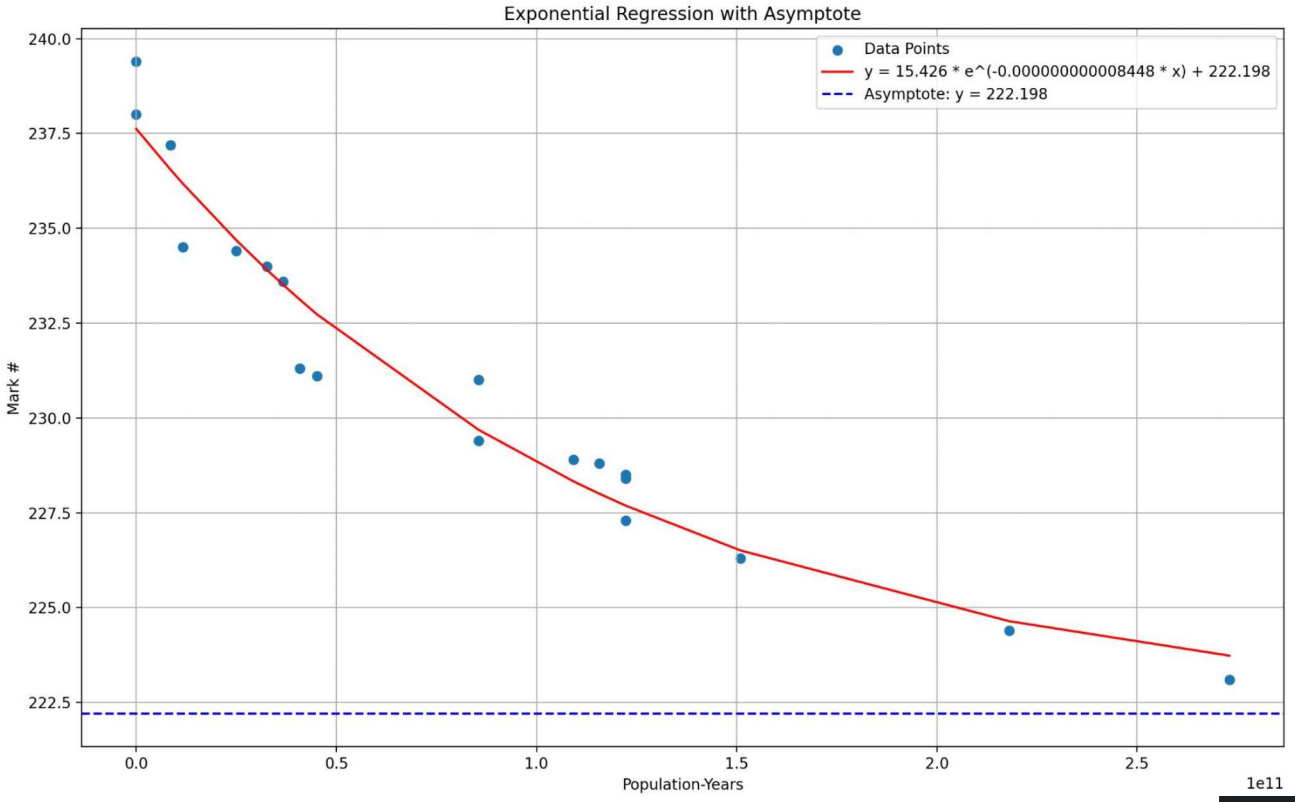

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Correlation coefficient: -0.9742
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Polynomial Regression World Records



Correlation coefficient: 0.9806

Exponential Regression with Asymptote World Record




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Correlation coefficient: 0.9761
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The fastest mile possible: 222.198 seconds
3:42.2 minutes
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Mathematical Modeling

- Using linear regression on world records → predicts impossible negative times
- Shows lack of adequate data
- A better model: population-years → exponential decay
- Yields predicted asymptote: **3:42.2**
- Only **0.93 seconds** faster than the current record

The Future of Human Performance

Human physiology sets real limitations
Performance relies on:

- VO2max
- Lactate threshold
- Running economy
- Neurological & genetic factors

Future improvements driven by:

- Scientific innovations
- Technology
- Training
- Doping

The Future of Human Performance

Jakob Ingebrigtsen, the 25-year-old Norwegian middle-distance star, says he's targeting the world records in the 1500m, mile, and 5000m next year (2026). Based on our exponential regression and population-weighted analysis, the theoretical fastest mile is 3:42.2 minutes.

While our model predicts an asymptotic limit, Ingebrigtsen's attempt could provide valuable real-world data to refine these projections and test how close elite athletes can approach human physiological limits.

Works Cited

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